



FACULTY OF ELECTRICAL-ELECTRONICS ENGINEERING
DEPARTMENT OF CONTROL AND AUTOMATION ENGINEERING
ADVANCED INDUSTRIAL AUTOMATION SYSTEMS APPLICATIONS
EXPERIMENT 3 – 4

OMRON SERVO DRIVE EXPERIMENT

There is two servo drive and one remote I/O module communicating with Ethercat. Your task is commanding these servo drives by using the input of the remote I/O module.

The working procedure

- Servo motors will be energized with the I1 button. When motors are energized, the Q1 lamp should light up and the Q2 lamp should light off.
- Servo motors will be de-energized with the I2 button. When motors are de-energized, the Q2 lamp should light up and the Q1 lamp should light off.
- Servo driver errors should be cleared with the I2 button.
- Both motors' velocities should be controlled with one analog input value.
- After you arrange the analog value, if you push the I3 button, the First motor should start turning at that speed. If you push the I4 button, the Second motor should start turning at that speed. If you push the I1 button, both motors should turn at that speed. If you push the I2 button both motors should stop.
- Q3 and Q4 lamps should light up according to the last input between I3 and I4. If you push I4, the Q4 lamp should light up and the Q3 lamp should light off.
- If you push I2 button, Q3 and Q4 lamp should light off. If you push the I1 button, the Q3 and Q4 lamps should light up.

PLcNEXT EXPERIMENT

Create a wave generator model using Simulink and add it as a user library to your PLCnext Engineer project. Wave generator model must satisfy these conditions:

- It should include only 2 inputs and 1 output, no other inputs and outputs are allowed.
 - First input should be all switches of the 1st module (dio-1 / ~DI8).
 - Second input should be the analog input (aio-1 / IN01).
 - The output should be the wave wanted to be generated.
- Depending on the switch activated/toggled, these signals should be generated.
 - Constant signal
 - Sawtooth signal
 - Amplitude = 15000
 - Frequency = 0.5 Hz
 - Sine signal
 - Amplitude = Bias = 15000
 - Frequency = 0.5 Hz
 - No signal when other cases
- Second input (analog) should act on the signals generated as follows
 - Direct output for constant signal
 - Bias for sawtooth signal
 - Bias for sine signal